2013 Q1

1. (a)	$(1.5 \times 1000\ \text{kg}) \times 4200\ \text{J kg}^{-1}\,°\text{C}^{-1} \times (80 - 60)\ °\text{C} \times (1-15\%)$ $= 1.07 \times 10^8\ \text{J}$	1M+1M 1A	3
(b)	$1.07 \times 10^8\ \text{J} \div (4.5\ \text{kW}) \div 3600\ \text{s}$ $= 6.61$ (hours)	1M 1A	2
(c)	Rate of heat transfer drops as the water temperature drops / the room temperature increases / temperature difference drops.	1A	1

2014 Q1

1. (a)	5 min. (or 300 s)	1A	1
(b)	When the heater is switched off, its temperature is higher than the metal.	1A	
	Or Heat continues to conduct to the metal until the same temperature is attained.	1A	
	Or It takes time for them to attain the same temperature.	1A	1
(c) (i)	$mc\Delta T = IVt$ $(0.80)\,c\,(45 - 20) = (4.0)(12)(5 \times 60)$	1M	
	$c = 720\ \text{J kg}^{-1}\,°\text{C}^{-1}$	1A	2
(ii)	Experimental result is higher than the actual value.	1A	
	Not all the energy supplied by the heater goes to the metal.	1A	
	Or Some energy is absorbed by the heater itself and / or by the thermometer.	1A	
	Or Heat loss to the surroundings.	1A	2
(d)	Glass is not a good conductor of heat.	1A	
	Or Prolonged time is required for the whole glass block to attain uniform temperature.	1A	
	Or No insulating material is perfect, so heat loss is appreciable.	1A	1

2016 Q1

1.	(a)	- Put the sphere into the water bath for a few minutes - Put / transfer the sphere into the polystyrene cup (of water) - Measure the final / maximum temperature T_f of the water with a thermometer	1A 1A 1A	
		$0.80 \times c_b \times (80 - T_f) = 0.50 \times 4200 \times (T_f - T_0)$ $c_b = 2625 \times \frac{T_f - T_0}{80 - T_f}$ (J kg^{-1} °C^{-1})	1A	
		Precautions: - Dry the sphere with the towel quickly before putting it into the cup - Make sure the sphere is fully immersed in water - Stir the water thoroughly Any **TWO**	1A 1A 1A	
			6	
	(b)	Thermal energy / heat is lost during the transfer / drying of the sphere Or Thermal energy / heat absorbed by thermometer, stirrer or cup Or The temperature of the sphere is higher than T_f when this final temperature is measured (i.e. T_f not reaching its maximum) Thus temperature rise of water in the cup is lower than it should be.	1A 1A 1A 1A	
			2	

2018 Q1

		Solution	Marks	Remarks
1.	(a)	Amount of energy required $E = mc\Delta T$ $= 6 \times 4200 \times (50 - 15)$ $= 882000$ J (or 882 kJ)	1M	
		Power $P = \frac{E}{t} = \frac{882000}{60}$ $= 14700$ W (or 14.7 kW)	1M 1A	
			3	
	(b)	Let m kg per minute be the water flow rate $mc\Delta T = Pt$ $m(4200)(40 - 15) = 14700 \times 60$ $m = 8.4$ (kg min^{-1} or kg)	1M 1A	Or $\frac{m}{t} \propto \frac{1}{\Delta T}$ $\therefore \frac{m}{6} = \frac{50-15}{40-15}$
			2	
2.	(a)	$n = \frac{pV}{RT} \propto pV$ (for constant T) $\frac{n_A}{n_B} = \frac{(p)(3V)}{(2p)(2V)}$ $n_A = 0.75 \times 0.8$ mol $= 0.6$ (mol)	1M 1A	

1. (a)	$5 \times 0.02 \times 3300 \times (T - 4) = 0.60 \times 4200 \times (96 - T)$ $T = 85.347368\ °C \approx 85.3\ °C$	1M 1A 2	Accept 85.0 °C to 85.4 °C [Marking Remark (MR): temperature difference must be correct & accept missing '5' – 1 M]
(b) (i)	To compensate for / balance the heat loss (of the container with soup) to the surroundings.	1A 1	[MR: Key words - heat loss to the surroundings (from container or through evaporation)]
(ii)	$P \times 10 \times 60 = 2000 \times 9 + 16 \times 4200 \times 9$ $P = 1038\ W \approx 1040\ W$	1M+1M 1A 3	Assume : Power of the heater = Rate of heat loss (of the container with soup) to the surroundings [MR: P×t ✓ + 1 term ✓ - 1M P×t ✓ + 2 term ✓ - 2M P×t: accept $P \times 10$ ΔT: accept (96 – 9)]
(iii)	Smaller than 9 °C. As the temperature of (the container with) soup drops, the rate of heat loss to the surroundings becomes lower due to a smaller temperature difference (w.r.t. the surroundings).	1A 1A 2	[2 A marks are independent Key concept: smaller temperature difference]

1.	(a)	Heat given out by iron cube = Heat gained by water $0.2 \times 450 \times (T - 33) = 0.6 \times 4200 \times (33 - 25)$ $T = 257$ (°C)	2M 1A	
			3	
	(b)	Some water might become steam (immediately). Or Some water in the cup might splash.	1A 1A	
			1	
	(c)	It is because some heat is lost to the surroundings through vaporization of water / as latent heat. Or It is because some heat is lost to the surroundings (or polystyrene cup) / The water might not be well stirred, otherwise, the water should have reached a temperature higher than 33 °C (i.e. underestimated).	1A 1A 1A 1A	
			2	
	(d)	(i) Poor thermal contact between the glass bob of the thermometer and the iron cube.	1A	
			1	
		(ii) Infra-red (IR) thermometer.	1A	
			1	

1.	(a)	The glass cup absorbs thermal energy (from the hot water) / Heat is lost to the glass cup.	1A	
			1	
	(b)	$4200 \times 0.4 \times (100 - 93) = H \times (93 - 23)$ $H = 168$ (J °C^{-1}) ≈ 170 (J °C^{-1})	1M 1A	
			2	
	(c)	(i) Infra-red (radiation) / thermal radiation	1A	
			1	
		(ii) Remote control / infra-red (IR) thermometer / night vision camera	1A	
			1	